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Drivers and Challenges of Micro-Irrigation Adoption and Intensity in India: A Study of Four States

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ABSTRACT

India's food security relies substantially on groundwater, but inefficient irrigation is seriously depleting this resource. Micro-irrigation (MI) improves irrigation application efficiency and productivity, yet adoption remains low. This study examines MI adoption across four major Indian states—Uttar Pradesh, Maharashtra, Madhya Pradesh and Telangana—using primary data, Heckman's two-stage regression and perception analysis. The sample is drawn from farmers with independent water sources, and the findings should be interpreted within this scope. The study identifies key determinants of adoption, including access to information, high-value crop cultivation, wild animal damage, landholding size and subsidies. High investment costs and energy requirements remain significant barriers, particularly for smallholders. Strengthening information dissemination, easing subsidy access, improving fencing and promoting land consolidation can boost MI adoption. Implementing these measures will improve irrigation efficiency and contribute to sustainable agriculture and long-term food security.

RÉSUMÉ

La sécurité alimentaire de l'Inde s'appuie considérablement sur les eaux souterraines, mais une irrigation inefficace épuise sérieusement cette ressource. La micro-irrigation (MI) améliore l'efficacité d'application de l'irrigation et la productivité, mais son adoption reste très faible. Cette étude examine l'adoption de la MI dans quatre grands États indiens, notamment l'Uttar Pradesh, le Maharashtra, le Madhya Pradesh et le Telangana, à l'aide de données primaires, d'une régression en deux étapes de Heckman et d'une analyse des perceptions. L'échantillon est. constitué d'agriculteurs disposant de sources d'eau indépendantes, et les résultats doivent être interprétés dans ce cadre. L'étude identifie les principaux déterminants de l'adoption, notamment l'accès à l'information, la culture de plantes à haute valeur ajoutée, les dégâts causés par les animaux sauvages, la taille des exploitations et les subventions. Les coûts d'investissement élevés et les besoins énergétiques constituent des obstacles importants, en particulier pour les petits exploitants. Le renforcement de la diffusion de l'information, la facilité de l'accès aux subventions, l'amélioration des clôtures et la promotion du remembrement des terres peuvent stimuler l'adoption de la MI. La mise en œuvre de ces mesures améliorera l'efficacité de l'irrigation et contribuera à une agriculture durable et à la sécurité alimentaire à long terme.

Article in French: Facteurs et défis de l'adoption et de l'intensité de la micro-irrigation en Inde: Une étude menée dans quatre États de l'Inde.

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1 | Introductionn

There is an imminent risk of 'water bankruptcy' globally (Madani 2026) and for India as it accounts for 23% of global groundwater depletion (Dangar et al. 2021). Groundwater supports over 60% of the country's agriculture, and extraction rates have surpassed natural replenishment and led to poor groundwater development (MoWR 2017; Bhattarai et al. 2021). The state of groundwater extraction in the key food-producing states—Haryana (134%), Punjab (166%), and Rajasthan (151%)—is significantly above the replenishment rates, while Madhya Pradesh (MP) (59%), Maharashtra (55%), Telangana (42%) and Uttar Pradesh (UP) (60%) are at margins of replenishment rates (CGWB 2023). The demand for freshwater is likely to swell further because of global megatrends: population growth, rapid urbanization, the emerging middle class and their increased affluence, growing inequality and climate change (Bates et al. 2008; King et al. 2017).

The irrigation application efficiency in Indian agriculture is estimated at approximately 35%–40% under canal-based systems and approximately 60% under groundwater-based irrigation at the field level (Chand et al. 2020). It is important to distinguish application-level efficiency from basin-level water use efficiency, which accounts for return flows and recycling. The case for MI adoption rests not only on field-level efficiency gains but also on the 'dry water savings' concept (Seckler 1996), that is, while return flows from surface irrigation may be recirculated within a basin, groundwater depletion from overextraction represents genuine nonrecoverable loss. MI reduces nonbeneficial evaporation and deep percolation, losses that cannot be recaptured, unlike surface return flows, while improving water control and beneficial evapotranspiration. However, these water-saving impacts are not uniform and are conditioned by the geo-hydrological environment, crop type, climate and MI technology deployed (Kumar 2016).

Micro-irrigation (MI) has been found to enhance returns from high-value crops by increasing yield and quality, reducing electricity consumption, increasing nitrogen use efficiency, presenting climate mitigation benefits, saving labour and adapting to varied terrains (Namara et al. 2005; Kumar 2016; Reddy 2016; Gandhi et al. 2021; Zhang et al. 2021; Pronti et al. 2024). Some research indicates a possible rebound effect in MI adoption, where efficiency gains may lead to an increase in absolute water use due to increased irrigated area (Scott et al. 2014; Song et al. 2018; Singh et al. 2024). Nevertheless, for the highly populous nation, enhancing agricultural productivity while managing depleting groundwater resources remains imperative, both to ensure the long-term sustainability of its food system and to safeguard national food security.

To accelerate the adoption of MI, the Indian government consolidated various schemes under *Pradhan Mantri Krishi Sinchayee Yojana* in 2015 with a budget of Rs. 500 billion ($5.95 billion) over 5 years. Despite significant policy efforts to promote MI, its adoption remains low. MI adoption is predominantly driven by the following: (a) the cultivation of high-value row crops (fruits, vegetables and flowers) where the return on MI investment is clearly positive, (b) farm-level water scarcity where conventional irrigation is severely constrained and (c) agro-ecological conditions favorable to drip/sprinkler application (Kumar, Turral, et al. 2008; Kumar 2016; Gandhi et al. 2021; Mohan et al. 2024). In cereal-dominated, canal-irrigated settings, the economic incentive to adopt MI is considerably weaker, partly explaining the persistent adoption gap in these states. The substantial spatial and agro-ecological heterogeneity in MI adoption across Indian states is presented in Table 1 and visually in Figure 1.

India has approximately 16.8 million hectares under MI, accounting for only 25% of the net irrigated area. Adoption is relatively high in states such as Karnataka (84%), Maharashtra (75%), Andhra Pradesh (66%), Mizoram (57%) and Tamil Nadu (49%). In contrast, large agrarian states that are central to national food security, including Punjab (2%), UP (3%), Bihar (4%), West Bengal (3%) and MP (9%), exhibit very low MI penetration relative to their extensive net irrigated areas. UP and MP together make an outsized contribution to India's agricultural output, accounting for approximately 31% of total foodgrain production, 37% of pulses and 27% of oilseeds. UP alone contributes nearly 47% of national sugarcane production.

National food and livelihood security remain heavily dependent on these regions, which continue to rely on inefficient and groundwater-intensive irrigation practices. The evidence highlights a dual challenge for India's MI strategy: extending adoption beyond eastern states where horticultural crops predominate and accelerating uptake in cereal-dominated, groundwater-stressed regions that rely extensively on subsidies for water, electricity and fertilizers (Ramaswami 2019; Government of India 2024). Between 1999–2000 and 2022–2023 (see Table 2), agricultural electricity consumption (a suitable proxy for water use) increased from approximately 91,000 GWh to approximately 244,000 GWh, representing a nearly 2.7 times rise with a compounded annual growth rate (CAGR) of 4.2%. During the same period, the gross irrigated area expanded from 63.2 to 122.3 million hectares or approximately 1.9 times (CAGR 2.8%) (Government of India 2024). The faster growth of electricity use relative to irrigated area indicates that irrigation expansion has been accompanied by increasing energy intensity, with electricity consumption per unit of gross irrigated area rising from approximately 1440 GWh/million ha in the early 2000s to nearly 1990 GWh/million ha in the early 2020s. Even after controlling for irrigation expansion, this increase in energy use per hectare points to deeper groundwater pumping, declining water tables or persistent inefficiencies in irrigation practices (Government of India 2024).

Our research paper is motivated by a gap in empirical and policy-relevant research in the literature on MI adoption. While prior studies on MI adoption have largely focused on relatively high-adoption states such as Gujarat, Maharashtra, Karnataka, Rajasthan and Tamil Nadu (Narayanamoorthy 2005; Kumar, Rajvanshi, and Dash 2008; Palanisami et al. 2011; Bhamoriya and Mathew 2014; Kumar 2016; Dar et al. 2017; Fishman et al. 2023; Mohan et al. 2024), large-scale econometric evidence from states with persistently low adoption, particularly UP and MP, remains scarce. Our study contributes by (1) providing Heckman two-stage estimates that separately identify adoption and intensity drivers, (2) combining econometric analysis with a structured perception survey based on the framework of Khandker and Gandhi (2018) and (3) focusing on states central

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TABLE 1 | Share of micro-irrigation adoption in comparison to net sown area and horticulture crops.

S. no.	State	Total MI area (Sprinkler + Drip)	Net irrigated area (NIA) ('000 ha)	MI as a share of NIA	Area under horticultural crop ('000 ha)	MI as a share of horticulture area
1	Andhra Pradesh	2095	3158	66%	1598	131%
2	Arunachal Pradesh	14	56	25%	42	33%
3	Assam	46	189	24%	663	7%
4	Bihar	133	3288	4%	1312	10%
5	Chhattisgarh	405	1438	28%	774	52%
6	Goa	28	24	118%	—	—
7	Gujarat	1946	5323	37%	2020	96%
8	Haryana	754	3492	22%	414	182%
9	Himachal Pradesh	15	121	12%	341	4%
10	Jammu and Kashmir	1	383	0%	341	0%
11	Jharkhand	61	247	25%	450	14%
12	Karnataka	2852	3412	84%	2684	106%
13	Kerala	36	311	11%	1593	2%
14	Madhya Pradesh	759	8422	9%	2604	29%
15	Maharashtra	2297	3058	75%	2291	100%
16	Manipur	16	36	43%	83	19%
17	Meghalaya	1	69	2%	154	1%
18	Mizoram	9	16	57%	158	6%
19	Nagaland	31	98	31%	148	21%
20	Odisha	229	1180	19%	1596	14%
21	Punjab	60	3936	2%	512	12%
22	Rajasthan	2507	7759	32%	1771	142%
23	Sikkim	24	10	245%	87	28%
24	Tamil Nadu	1455	2954	49%	1618	90%
25	Telangana	359	1577	23%	404	89%
26	Tripura	7	81	8%	146	5%
27	Uttar Pradesh	430	13,703	3%	2608	16%
28	Uttarakhand	35	334	10%	165	21%
29	West Bengal	157	3428	5%	2087	7%
30	Total	16,761	68,103	25%	28,665	58%

Source: Agricultural statistics at a glance 2024.

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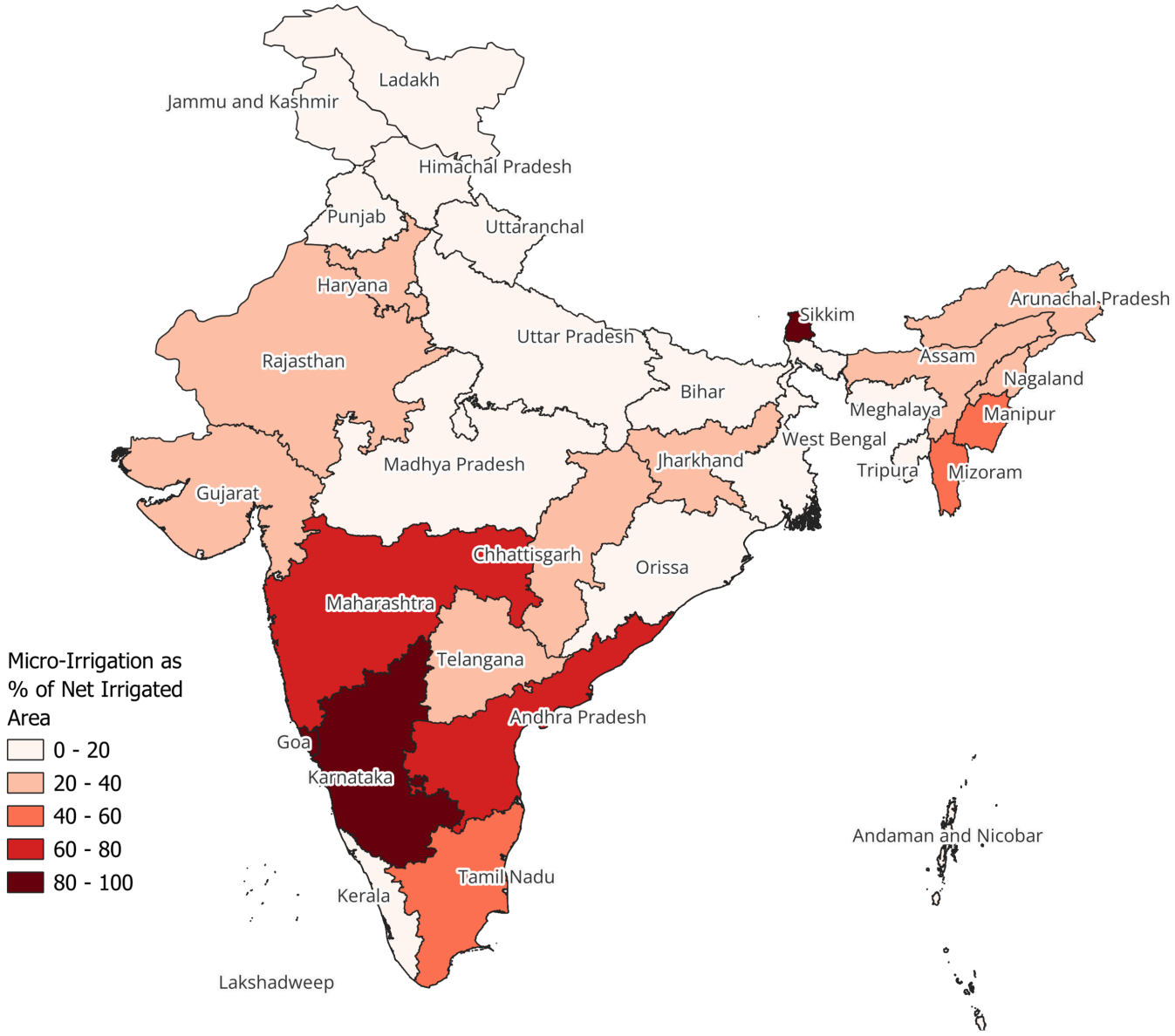


FIGURE 1 | Micro-irrigation share of net irrigated area of India.

to national food security but underrepresented in the MI adoption literature.

2 | Materials and Methods

2.1 | Sampling Strategy

The study collected data from four Indian states: UP, MP, Maharashtra and Telangana (see Figure 2). These states were chosen for four key reasons: (1) UP and MP exhibit a significant gap between MI potential and adoption (E&S DAC 2018); (2) UP, MP and Maharashtra hold the highest groundwater potential, contributing 16%, 8% and 7% of the national total, respectively (CWC 2019); (3) Maharashtra and UP experience high groundwater depletion rates and are the largest producers of sugarcane and (4) the selection includes both low-adoption states (UP and MP) and high-adoption states (Maharashtra and Telangana), ensuring a balanced perspective (see Table 1).

Given the need to find nearby adopters and nonadopters in some diverse settings, purposive sampling was adopted to focus on farmers who had adopted MI using subsidies. Figure 3 outlines the purposive sampling strategy. We selected two districts in each state based on MI coverage, groundwater stress and water-intensive crop cultivation (e.g., sugarcane, banana, wheat and cotton). The information was gathered from state agriculture departments. Within each district, two blocks and two villages per block were chosen on similar principles. In each village, farmers were randomly sampled, ensuring representation across landholding categories. The sampled adopter farmers had generally used MI since 2016, and nonadopters were sampled from the same villages for comparison.

2.2 | Data

Agricultural research often requires long-term data, but retrospective recall is a practical yet imperfect option due to

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TABLE 2 | Yearly electricity consumption and change in gross irrigated area in India.

Year	Electricity consumption for agricultural purposes (GWh)	Gross irrigated area (Mha)
1999–2000	90,934	63.2
2000–2001	84,729	76.19
2001–2002	81,673	78.37
2002–2003	84,486	73.06
2003–2004	87,089	78
2004–2005	88,555	81.08
2005–2006	90,292	84.28
2006–2007	99,023	86.75
2007–2008	104,182	88.06
2008–2009	107,776	88.9
2009–2010	119,492	85.09
2010–2011	126,377	89.32
2011–2012	140,960	91.93
2012–2013	147,462	92.78
2013–2014	152,744	96.27
2014–2015	168,913	97.85
2015–2016	173,185	97.75
2016–2017	191,151	99.44
2017–2018	199,247	101.27
2018–2019	213,409	104.49
2019–2020	211,295	112.23
2020–2021	221,303	118.93
2021–2022	228,451	120.38
2022–2023	243,852	122.29

Source: Agricultural statistics at a glance 2024.

potential recall errors (de Nicola and Giné 2014). Despite risks such as telescopic errors and false reporting, recall data are widely used in decision science and agricultural studies and often show minimal bias over time (Beegle et al. 2012; Singh et al. 2024). To ensure data completeness and reliability, we developed and piloted a survey in Gujarat, UP and Maharashtra before refining it for broader use. Native speakers from Agro-Economic Research Centres in Pune, Jabalpur, Allahabad and Visakhapatnam conducted the survey across Maharashtra, MP, UP and Telangana, respectively. The survey schedule was administered from December 2019 to March 2020 and covered farmer demographics, land use, irrigation sources, cropping patterns and MI adoption challenges, referencing the 2018–2019 crop year. The final dataset included 404 MI adopters (138 sprinklers, 264 drips) and 97 nonadopters from 86 villages in eight districts collected just before the COVID-19 lockdowns (Table 3). These farmers grew one or more water-intensive crops. Farmers had more than one crop under MI, which is why the total numbers in Table 4 are greater than the actual sample size of 404 adopters.

2.3 | Method of Analysis

We employed two analytical approaches to examine MI adoption data: the Heckman two-stage regression and a perception analysis. To avoid selection bias of the data, we adopted the Heckman two-step model. The Heckman model first estimates adoption probability using a probit regression (Equation 1) and then applies ordinary least squares (OLS) to analyse technology intensity (Equation (2)). The inverse Mills ratio (IMR) corrects for selection bias, ensuring robust estimation. Key explanatory variables include farmer demographics, land characteristics, access to credit, subsidies and information. Prior to running the model, we checked for multicollinearity, normality and homoskedasticity.

$$\Pr\left(Y_i = 1 \mid \mathrm{X}_i\right) = \Phi\left(X_i\,\beta + \epsilon_i\right) \tag{1}$$

where Y_i is a binary indicator of technology adoption; X_i is a vector of observed household, farm, institutional and agro-ecological characteristics; β is a vector of parameters; and Φ denotes the standard normal cumulative distribution function

$$\textit{Technology Intensity}_i = Z_i\boldsymbol{\gamma} + \delta\boldsymbol{\lambda}_i + \mu_i \tag{2}$$

where Z_i is a vector of explanatory variables influencing technology intensity, γ is a vector of parameters, λ_i is the IMR obtained from the first-stage selection equation, δ (delta) captures the correction for sample selection bias and μ is the error term.

In the second approach, the perception analysis assessed adoption drivers and barriers using responses on a 5-point Likert scale, where 1 = *strongly disagree*, 2 = d*isagree*, 3 = *partially agree/disagree*, 4 = *agree* and 5 = *strongly agree*. Farmers responded to a conceptual framework and challenge-based questions adapted from Khandker and Gandhi (2018) and our pilot survey. Table 5 presents some sample questions. Questions pertain to yield benefits, cost reduction, credit access and market challenges. Responses, analysed in R's 'Likert' package, provided graphs highlighting key perceived constraints.

2.4 | Descriptive Statistics of the Variables

Tables S1 and S2 present the descriptive statistics for numerical and categorical variables, respectively, across key dimensions of the sample, including socio-economic, institutional, farm structure, biophysical and farm management characteristics. In both tables, we have also indicated the meaning of each variable and referred source. Adopters of MI technology tend to have larger farm sizes (mean = 2.4 ha) than nonadopters (mean = 2.2 ha), indicating that medium and large farmers may be more likely to adopt MI technology. Adopters exhibit slightly higher levels of education (mean = 3.78 years) than nonadopters (mean = 3.41 years), suggesting that better-educated farmers may be more inclined to embrace modern irrigation methods.

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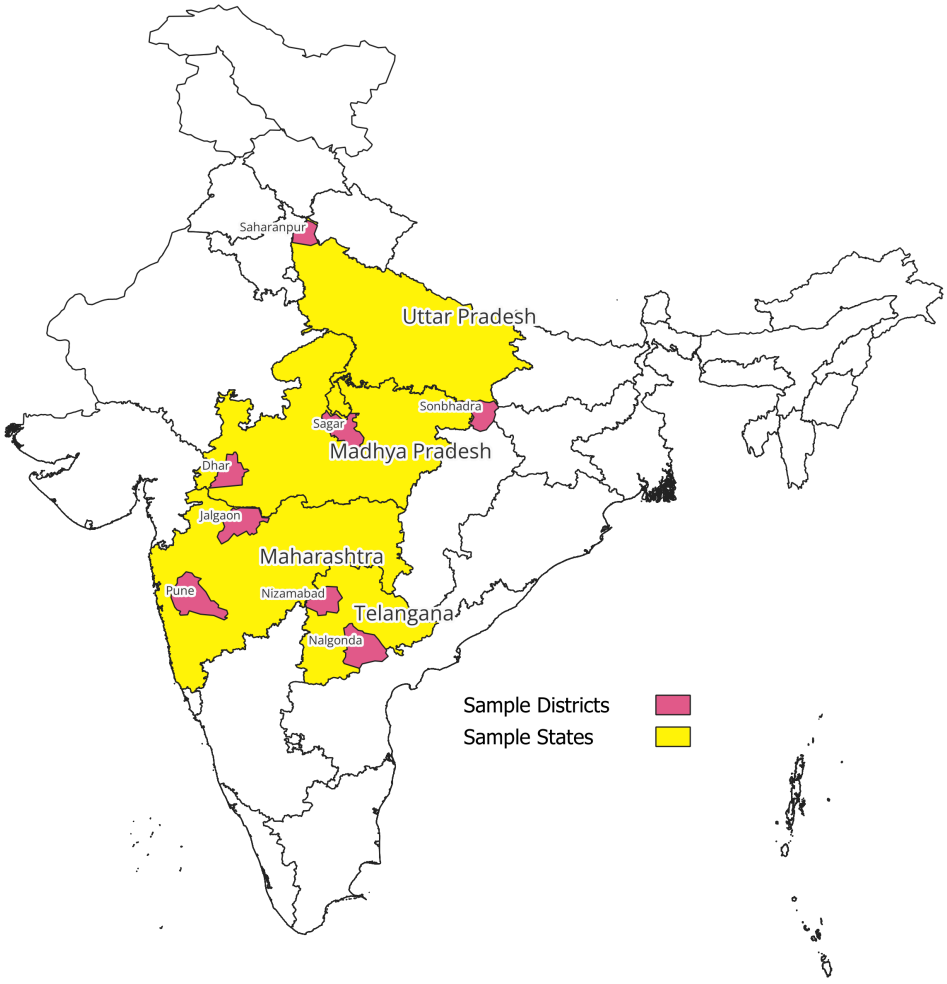


FIGURE 2 | Geographic location of sampling states and districts.

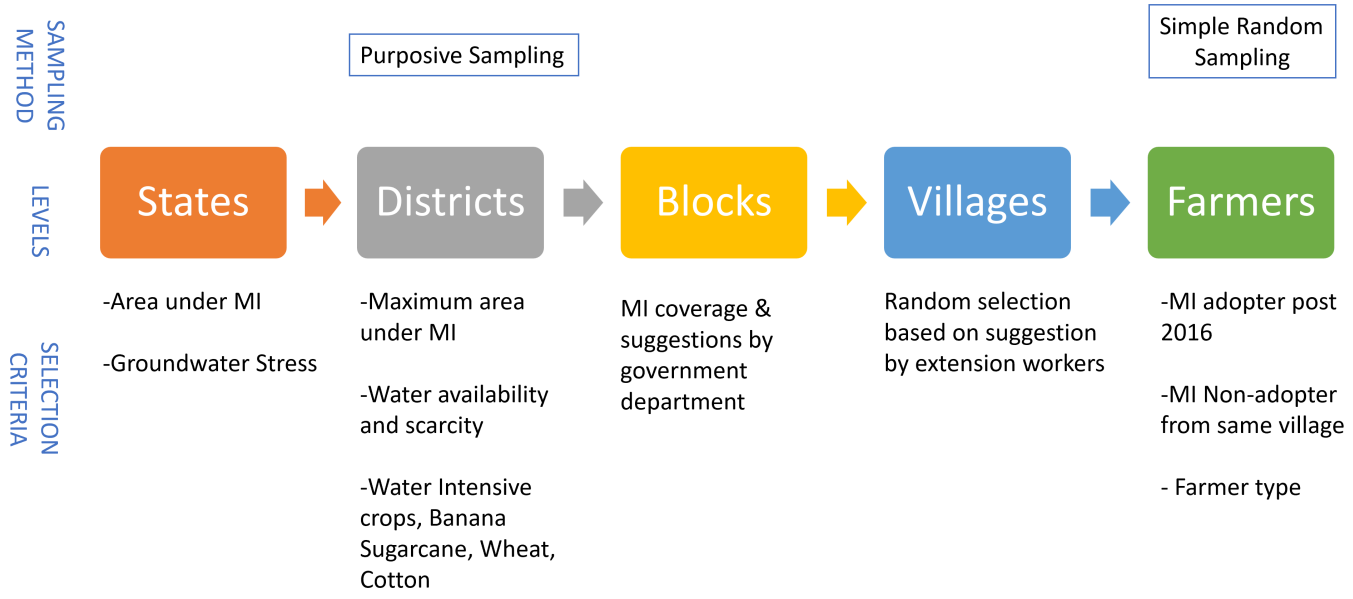


FIGURE 3 | Data sampling strategy.

Adopters report easier access to credit (mean = 3.32) and information (mean = 4.20) compared to nonadopters, who score lower on these factors (mean = 2.91 for credit and 3.28 for information). Fifty-eight percent of the farmers had tubewells as their source of irrigation. We have listed the literature and rationale for the relevance of these variables in both tables.

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TABLE 3 | Distribution of data among sampled districts.

S. no.	District surveyed	No. of village	Total adopters	Adopters—drip	Adopters sprinkler	Adopters—both	Nonadopters
1	Sonbhadra	6	48	16	32	—	12
2	Saharanpur	7	48	28	20	—	12
3	Dhar	6	48	48	—	—	12
4	Sagar	17	48	—	48	—	12
5	Pune	14	52	51	—	1	12
6	Jalgaon	19	64	64	—	—	13
7	Nizamabad	7	48	9	38	1	12
8	Nalgonda	10	48	48	—	—	12
	Total	86	404	264	138	2	97

Source: Primary data, authors' compilation.

TABLE 4 | State-wise and crop-wise distribution of sample farmers surveyed for various crops.

Crops	Uttar Pradesh	Madhya Pradesh	Maharashtra	Telangana	Total
Wheat	51	48			99
Sugarcane	30		52		82
Chickpea		36		35	71
Cotton			28	40	68
Banana		41	9		50
Chilli	12	30			42
Soybean				38	38
Total	93	155	89	113	450

Note: The sampled farmers also grew other high-value crops under MI.

TABLE 5 | Examples of perception questions.

S. no.	Examples of perception questions
1	**Conceptual framework questions (Khandker and Gandhi 2018)**
1.1	Micro-irrigation increases yield/output
1.2	Micro-irrigation reduces input use/costs
1.3	Information on micro-irrigation is easily available
1.4	There are a large number of companies supplying micro-irrigation equipment
1.5	The dealers provide good quality products you can trust
2	**Major problems faced by farmers**
2.1	Damage by animals
2.2	Difficulty in obtaining government subsidy and support
2.3	Lack of micro-irrigation dealers in area
2.4	High cost of maintenance in micro-irrigation

3 | Analysis

The analysis is divided into three sections: factors influencing adoption, adoption intensity and perceived challenges. The Heckman two-stage results (see Table S3) confirm selection bias, which, if unaccounted for, could distort technology intensity estimates. A rho value of −0.865 ($p = 0.000$) indicates a strong negative correlation between selection and outcome errors, reinforcing the need for correction. The Wald test ($\chi^2 = 25.85$, $p < 0.001$) rejects the null hypothesis of model independence, and this justifies the two-stage Heckman approach.

3.1 | Determinants of MI Adoption: First-Stage Probit Regression Analysis

The result of the first-stage selection model identified some key determinants of MI adoption. The findings are presented in Table S3. We find that access to information ($\beta = 0.596$, $p < 0.001$) and education ($\beta = 0.105$, $p = 0.029$) significantly increase adoption likelihood, indicating that more informed and educated farmers may be better equipped to evaluate and adopt new technologies. The positive association between land fragmentation and adoption ($\beta = 0.208$, $p = 0.004$) contrasts with the

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TABLE 6 | Farm size distribution of operated land, MI area and technology intensity.

Farmer category	Mean operated land (hectares)	Mean MI area (hectares)	Mean technology intensity (%)
Marginal	0.82	0.76	94%
Small	1.48	1.23	84%
Medium	4.03	2.86	74%
Large	13.35	8.15	57%
Grand total	3.16	2.27	80%

general finding that larger, consolidated holdings drive MI adoption (Kumar, Rajvanshi, and Dash 2008; Ruzzante et al. 2021). This reflects our sample's skew toward horticultural growers in Maharashtra and Telangana, where smallholder plot-level adoption is common and should be interpreted with caution in this context. We acknowledge that fragmentation and farm size are correlated; the positive association, after controlling for farm size, likely reflects efficiency-seeking behavior among farmers managing scattered plots rather than fragmentation as an independent structural driver. The positive association of the Herfindahl index with adoption ($\beta = 0.820$, $p = 0.022$) likely reflects the indirect effect of high-value crop cultivation. Farmers cultivating high-value row crops—banana, vegetables, and sugarcane—are more likely to adopt the technology (Dhawan 2000; Kumar 2016). These crops are agronomically amenable to drip and sprinkler systems and offer a strong return on MI investment. Farmers relying on tubewells showed higher adoption rates ($\beta = 0.5015$, $p = 0.020$), as MI aligns well with tubewell-based irrigation compared to other sources, although evidence of adoption under canal irrigation with storage systems (*diggie*) also exists (Kumar, Rajvanshi, and Dash 2008). The positive association between heavy (clayey) soils and adoption ($\beta = 0.9458$, $p = 0.010$) contrasts with much of the extant literature, which identifies light (sandy) soils as a more consistent driver of MI adoption due to higher percolation losses under conventional irrigation (Dhawan 2000). This discrepancy may reflect crop–soil interactions specific to our sample, where waterlogging risk in clayey soils creates incentives for precise water application in high-value crops such as sugarcane and banana. Farmers with larger landholdings are more likely to adopt MI, as indicated by significant coefficients for small ($\beta = 0.3977$, $p = 0.047$), medium ($\beta = 0.6159$, $p = 0.002$) and large farmers ($\beta = 1.476$, $p = 0.001$) compared to marginal farmers. Other variables, including age, credit access, subsidies, household size, market distance and water conditions, were not found to be statistically significant, indicating no major association with MI adoption.

3.2 | Second-Stage OLS: Factors Influencing Technology Intensity

In the second-stage regression analysis, we examined factors influencing technology intensity, defined as the percentage of farmland brought under MI. Key determinants included crop concentration, farmer size, land fragmentation and state-wise variations. The Herfindahl index ($\beta = 20.49$, $p = 0.002$) showed that farmers with concentrated cropping patterns expanded MI more, likely because such specialization often involves high-value crops supported by established market linkages. Conversely, land fragmentation ($\beta = -2.19$, $p = 0.022$), which initially appeared to encourage adoption, was found to hinder intensive use and expansion due to water supply, road and electricity access constraints in fragmented holdings. Market distance ($\beta = -0.2287$, $p = 0.094$) was negatively associated with expansion, as proximity to markets facilitates input access and output sales. Heavy soils ($\beta = -15.19$, $p = 0.08$), though encouraged adoption, may limit expansion. Larger farms exhibited lower MI intensity, as indicated by the coefficients of large ($\beta = -35.04$, $p < 0.001$), medium ($\beta = -13.59$, $p < 0.001$), and small farmers ($\beta = -8.13$, $p = 0.003$). The lower proportional MI intensity among large farmers ($\beta_large = -35.04$, $p < 0.001$) reflects a denominator effect, as total operated land increases the share under MI declines even though the absolute area under MI rises. Table 6 shows that the mean MI area increases steadily with farm size from 0.76 ha for marginal farmers to 8.15 ha for large farmers, while the proportional intensity falls from 94% to 57%. Therefore, this pattern does not indicate lower adoption among large farmers in absolute terms but rather a scaling effect where MI occupies a smaller proportion of a much larger operational base. State-level factors mattered, with Maharashtra ($\beta = 36.26$, $p < 0.001$) and Telangana ($\beta = 34.93$, $p < 0.001$) showing higher adoption intensity than UP, likely due to stronger institutional support and greater water scarcity in these states.

3.3 | Farmers' Perception on the Benefits of MI Adoption

Table 7 reports farmers' perceptions of MI potential, effective demand and supply. The results indicate that agronomic potential plays a key role in MI adoption. As per total agreement % in Table 7, all respondents agreed that MI improves yield, while 99% acknowledged water savings, 68% acknowledged labour reduction and 57% acknowledged fertilizer savings. Better weed control under MI was agreed by 75%, and 43% agreed on lower incidence of pests after MI adoption. These findings reinforce the technology's agronomic benefits as a major driver. The results showed that 94% believed MI increases profitability, 92% saw improved product quality and 63% noted reduced input costs. While 90% valued subsidies, 35% found capital costs of MI as high, indicating affordability concerns for some despite expected returns. In effective demand factors, 92% considered the technology easy to use, and 90% confirmed good information availability on MI. However, water availability and electricity reliability (71% and 62%, respectively) were indicated as important concerns. In terms of supply and distribution factors,

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TABLE 7 | Distribution of perception responses on agronomic, agro-economic, effect demand and actual consumption.

Variable/statement	Strongly disagree	Disagree	Neutral	Agree	Strongly agree	Total agreement
Agronomic potential						
Increases yield/output	0%	0%	0%	62%	38%	100%
Saves water	0%	0%	1%	51%	48%	99%
Reduces fertilizer use	2%	11%	30%	41%	16%	57%
Reduces pest/pesticide use	3%	13%	42%	35%	7%	43%
Reduces weed problem	2%	9%	15%	63%	12%	75%
Reduces labour use	2%	13%	17%	52%	16%	68%
Agro-ecoonomic potential						
Capital cost not high	6%	29%	16%	37%	12%	49%
Raises output quality/profit	0%	0%	8%	65%	27%	92%
Reduces input/costs	0%	16%	21%	44%	20%	63%
Increases profitability	0%	0%	6%	57%	37%	94%
Subsidy is substantial	0%	3%	6%	52%	38%	90%
Effective demand						
Info easily available	0%	3%	7%	56%	33%	90%
Technology easy to use	0%	1%	7%	63%	29%	92%
Subsidy easy to get	4%	19%	19%	48%	10%	58%
Finance easy to get	3%	25%	20%	45%	8%	54%
Electricity available/reliable	14%	10%	14%	48%	14%	62%
Water supply sufficient	3%	10%	16%	54%	17%	71%
Actual use and supply						
Many companies supply equip.	2%	13%	12%	57%	16%	73%
Equipment quality good	2%	5%	15%	59%	19%	78%
Dealers located nearby	4%	21%	11%	44%	21%	65%
Dealers: good products	1%	6%	13%	64%	16%	80%
Dealers: reasonable price	2%	11%	25%	54%	9%	62%
Dealers arrange subsidy/credit	10%	16%	10%	45%	19%	64%
Dealers: after-sales service	8%	17%	17%	45%	14%	59%

Note: Total agreement % is sum of agreed and strongly agreed % values. The colour in green to yellow, highlights the high or low percentage of the total agreement, where green colour signifies high percentage while yellow presents low percentage.

80% found product quality to be satisfactory, but confidence in after-sales support and pricing was lower, suggesting areas for improvement.

3.4 | Adopters and Nonadopter Farmers' Perspectives on Challenges in the Adoption of MI

Figure 4 highlights the results on key challenges in adopting MI. The top concerns of farmers were damage to MI laterals and crops from wild animals (64%), lack of fencing (60%), which could prevent animal intrusion. Animals such as wild boars, blue bulls, squirrels and rats are reported to frequently damage MI, causing losses and crop failure, making fencing important for adoption. A declining groundwater table (45%) is also an impending challenge, alongside high well and tubewell installation costs (43%), limited credit access (40%) and difficulties in obtaining subsidies (39%). As farmers pump water from deeper levels, costs rise because larger pumps are needed, and high-TDS water can clog laterals when drip lines dry in the sun (Kumar 2016), discouraging adoption.

Figure 5 presents findings on hurdles perceived by nonadopters in the adoption of MI. The most cited barrier to MI adoption was high investment cost, with 61% of respondents identifying it as a major deterrent. About 51% of nonadopter farmers reported that

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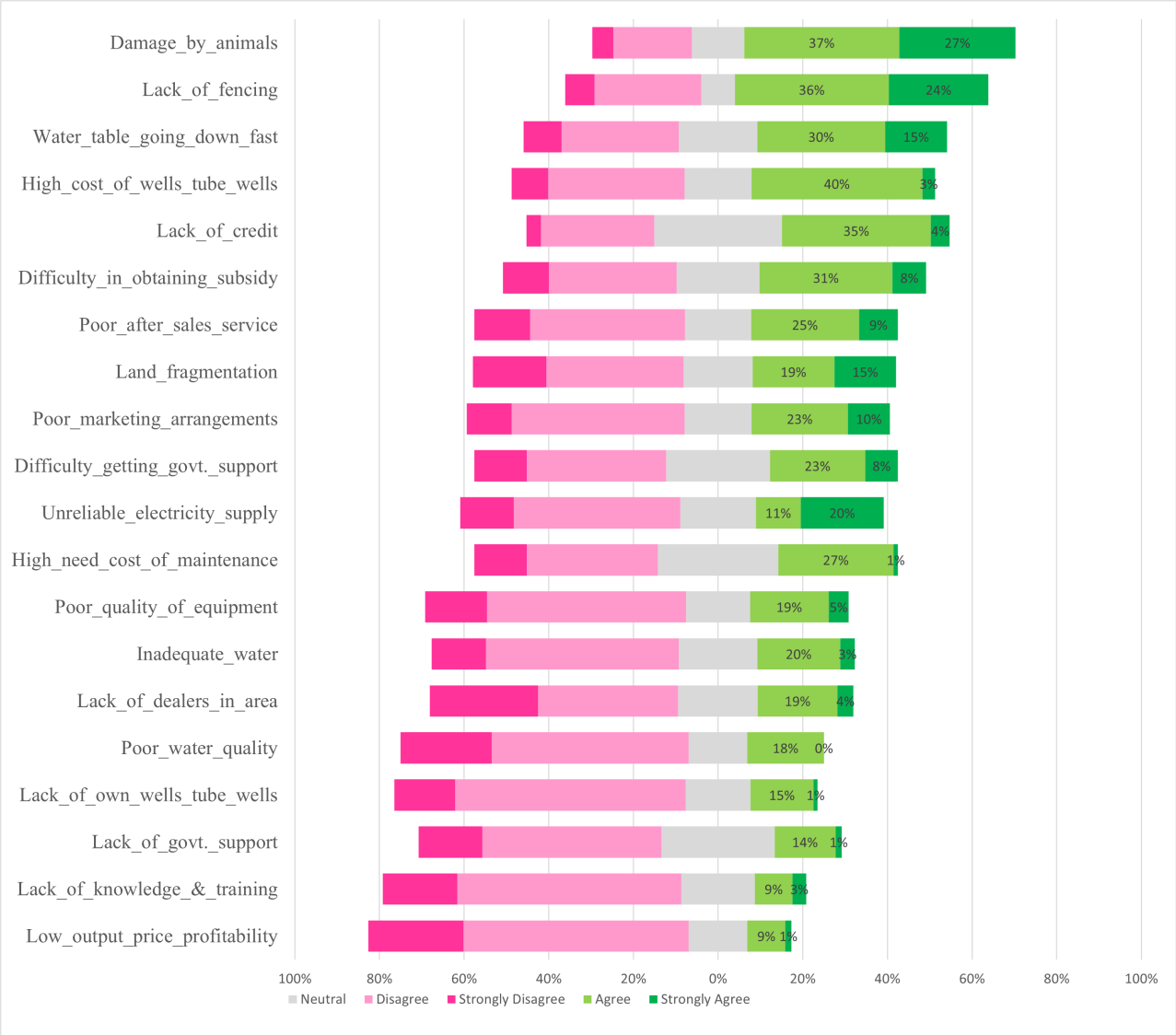


FIGURE 4 | Perceived challenges of adopters in MI adoption. *Source:* Authors' compilation using primary data.

subsidies for MI adoption are insufficient, while 45% cited high operating costs as another constraint. Difficulty in accessing credit (44%) and concerns about the risk of crop damage by animals (42%) were also noted as significant barriers. Despite these challenges, 71% disagreed that MI is unprofitable, suggesting that farmers recognize its benefits but lack investment resources and consider it risky, given the cost of adoption. To encourage MI adoption, policies should focus on support and risk reduction strategies, such as improved access to credit, easy access to subsidies, support for fencing and crop insurance. Extension services providing training in rodent and pest management could also enhance adoption rates.

An analysis of responses to open-ended questions regarding challenges revealed that the most frequently cited concern among farmers (45% of respondents) was damage to drip pipes caused by animals, including rats, wild pigs, squirrels and dogs. The second most commonly reported issue was the clogging of pipes and nozzles, primarily due to poor water quality. With respect to policy recommendations, approximately 50% of respondents prioritized easing subsidy access and reducing the Goods and Services Tax on MI equipment. The second most prominent suggestion pertained to the provision of fencing support as a measure to mitigate animal-induced damage, followed by calls for timely subsidy disbursement and simplified administrative processes.

4 | Discussion

4.1 | Factors Determining the Adoption of MI

MI adoption is shaped by interacting physical, socio-economic and institutional constraints spanning subsidies, finance and extension (Kumar, Turral, et al. 2008). Information access is central to this, as the source and type of information influence not only technology awareness but also access to credit, subsidies and maintenance services (Ruzzante et al. 2021). While private input suppliers accelerate adoption in India, they often embed rent-seeking motives that distort information flows (Malik et al. 2018; Dar et al. 2024). Reliable information reduces asymmetries and enables adoption, although its impact is mediated by market failures and source credibility (Adegbola

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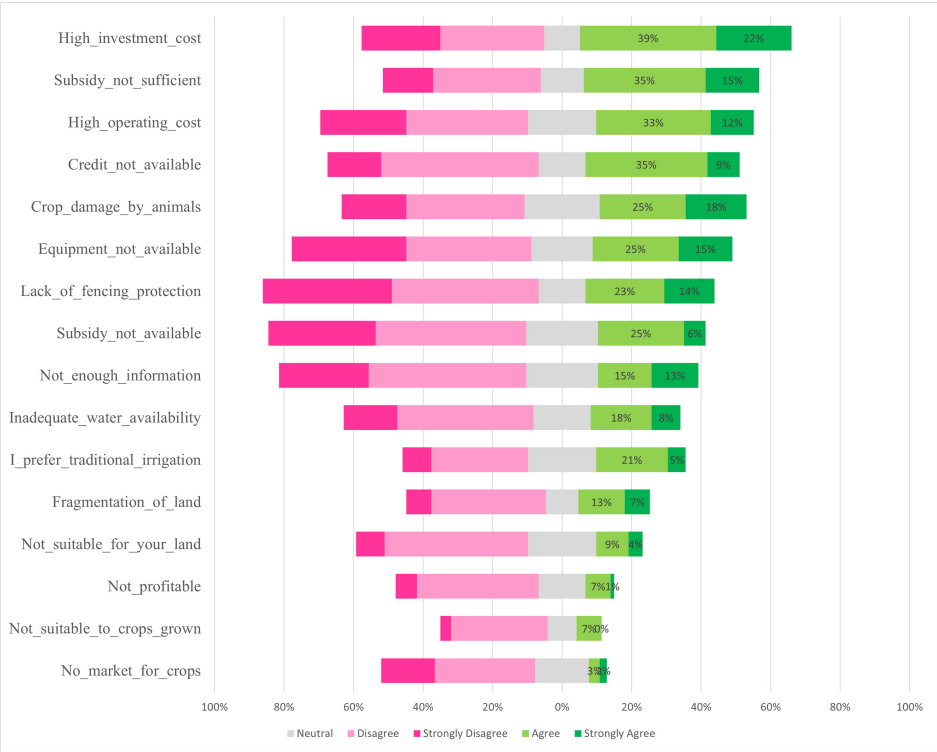


FIGURE 5 | Perceived challenges of nonadopters in micro-irrigation adoption.

and Gardebroek 2007; Aker et al. 2016). Education shapes how individuals process and act on such information (Zulfiqar and Thapa 2018). Since expanding formal education is a long-term objective, targeted extension campaigns on technology literacy, credit, insurance and subsidy access through village institutions offer a more immediate pathway to improving adoption outcomes.

Larger landholdings may involve more fragmented parcels, but they also increase farmers' capacity to invest regardless of fragmentation. Evidence shows that farmers with larger landholdings are more likely to adopt new technologies (Madramootoo and Morrison 2013; Ruzzante et al. 2021). They have better financial resources, access to credit and extension and a greater ability to manage risk, which supports adoption. While land fragmentation complicates irrigation management, large farmers perceive the benefits of MI—profitability, cost reduction, and increased yields—as outweighing these challenges. The positive association of the Herfindahl index with adoption reflects farmers' tendency to concentrate on high-value crops amenable to MI rather than monoculture per se (Kumar, Turral, et al. 2008; Fishman et al. 2023). Crop type and value are the underlying drivers of adoption; the Herfindahl index captures this indirectly, as farmers growing such crops naturally tend toward concentration.

The study finds adoption across all farm sizes, with higher intensity among small farms. The lower proportional intensity among large farmers reflects a denominator effect, where the share under MI declines as total operated land increases, even though the absolute area continues to rise. Our research confirms that groundwater-dependent farmers show a greater propensity to adopt MI than those relying on other sources, unsurprisingly, since drips and sprinklers are poorly suited to canal irrigation. The mismatch between surface water delivery schedules (once every 10–15 days) and MI's requirement for frequent, low-volume applications necessitates costly intermediate storage and pumping infrastructure, undermining economic viability (Kumar 2016). Even among groundwater users, adoption remains constrained where farmers depend on water purchase or where poor groundwater quality, as varying total dissolved solids levels reaching up to 2000 ppm in parts of India, causes dripper clogging and system failure. While expanding tubewell installation is inadvisable, any new groundwater source should be accompanied by policies that actively encourage water conservation and MI adoption. Farmers adopt more in heavy soils, where MI reduces waterlogging, improves input efficiency and increases yields (Chen et al. 2002).

Our perception analysis confirms that agronomic and agroeconomic benefits drive MI adoption, consistent with prior research (Bansal et al. 2021; Gandhi et al. 2021; Zhang et al. 2021). While all farmers acknowledged yield improvements, both adopters and nonadopters raised concerns about high investment and operating costs, insufficient subsidies, limited credit access and wildlife damage to drip systems, a problem compounded by inadequate fencing. Broader structural concerns, including declining groundwater, deteriorating quality and rising tubewell costs, further temper enthusiasm for adoption.

4.2 | Role of Subsidies

Access to subsidies did not emerge as significant in the Heckman two-stage model, contrasting with prior research (Yao et al. 2017; Porteous 2020; Ruzzante et al. 2021) and

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farmers' perception in our study. This is partly because MI can be economically viable without subsidies in some settings (Kumar 2016) and because high transaction costs in accessing them may offset their value (Nair and Thomas 2023). Subsidies are typically accessed through innovation intermediaries who are essential to technology distribution but may extract excessive rents, inflating effective technology prices (Malik et al. 2018; Nair and Thomas 2023). Poor administration worsens this, as direct government payments to manufacturers lead to neglected after-sales services and substandard systems, while influential farmers capture most benefits at the expense of marginal ones, and manufacturers inflate prices to offset documentation costs, undermining the subsidy's intended welfare impact (Kumar, Turral, et al. 2008).

Retailers act as rent-seeking agents exploiting brokerage opportunities between farmers and bureaucratic subsidy programs, often through credit-input-lock mechanisms requiring farmers to purchase inputs and sell produce to repay loans at high interest rates (Sidhu and Gill 2006; Malik et al. 2018). In the four study states, MI is one of many functions within agriculture or horticulture departments, limiting focus and accountability. Improving subsidy administration and reducing transaction costs to access subsidies are therefore critical. The Gujarat Green Revolution Company (GGRC) offers a strong institutional model, ensuring transparent support, third-party inspection and after-sale services, which have improved adoption and smallholder inclusion (Kumar, Turral, et al. 2008; Liebrand 2017).

4.3 | Scaling MI

Analysing technology adoption intensity reveals the impact of factors such as smaller fragmented plots, soil type, market accessibility and farm size on the adoption of MI. Farmers cultivating a single high-value crop are more likely to expand MI coverage due to economies of scale and comparative advantage over cereal crops. Conversely, farmers with smaller plots face compounding infrastructure constraints, such as a lack of independent water sources, electricity and road access, that hinder scaling, often pushing them toward water markets (Kumar, Turral, et al. 2008). These findings echo broader structural barriers to MI adoption—the dominance of small and marginal farmers, mismatched water delivery schedules in canal commands, poor groundwater quality in arid regions and weak extension—and subsidy administration collectively limits uptake beyond well-endowed, commercially oriented farms. We learned that farmers are also cautious about scaling due to potential crop damage from animals such as blue bulls, boars, stray cattle and rats. Adoption of MI would require additional investments in wire fencing to protect crops, leading farmers to assess the cost–benefit carefully. Market access also influences whether a farmer will expand their area, as they may choose to expand for crops that can be easily sold. Farmers tend to prefer high-value crops, such as vegetables and fruits, which rely heavily on nearby markets due to their perishability (Fishman et al. 2023). This aligns with evidence that drip irrigation is most economically viable for high-value horticultural crops and orchards, where even marginal yield gains translate into significant revenue increases and quality improvements fetch better market prices (Kumar 2016; Dhawan 2000). Farmers' responses about soil type affecting adoption may pertain only to the specific fields where MI is applied, rather than covering diverse soil conditions across their entire holdings.

4.4 | Policy Implications

The study strongly emphasizes the need for reliable information dissemination, including through digital and social media channels, to reduce information asymmetry about the technology and the process of subsidies. While fragmented landholdings do not deter farmers from adopting technology, they limit the expansion of area due to scaling challenges. Therefore, the government might consider emphasizing land consolidation policies to enable fragmented land to be exchanged among farmers. Dedicated institutional mandates for MI, similar to GGRC's model, should be established to improve focus and accountability. Performance-linked retailer payments should be adopted to reduce rent-seeking and eliminate credit-input lock practices, ensuring farmers pay only the subsidized share upfront and retailers are reimbursed directly. Crop damage caused by wild boars, blue bulls, rodents and abandoned cattle significantly influences farmers' technology adoption decisions and crop choices. Wild animals increase the risks associated with investing in fixed-field technologies such as MI. Policies promoting wire or live fencing, crop insurance, rodent management extension and ensuring the management of stray unproductive cattle in shelters could significantly reduce this risk. Efforts to increase the adoption of wire/live fencing should be encouraged, along with support for entrepreneurs who supply and maintain these fences.

4.5 | Limitations and Extension of Study

The study relies on recall data from adopters and nonadopters, which may affect response accuracy. Land fragmentation was measured using a Likert scale, although an index such as the Simpson Index could allow for clearer analysis. Fragmentation is largely shaped by inheritance and demographic factors; therefore, reverse causality is unlikely, but omitted variables such as land markets or local irrigation norms may bias estimates. The Heckman approach corrects for sample selection but does not fully resolve endogeneity. Future work using instrumental variables or panel data would strengthen causal inference. Water scarcity was captured through a perception-based ordinal proxy rather than a quantitative measure such as season-wise irrigated area. Future studies should collect season-specific irrigation data to build stronger farm-level indicators. The purposive sample includes mainly subsidy beneficiaries with independent irrigation sources, excluding those who buy water, share wells or depend on surface systems, so findings should be read within this scope. Network effects were not captured, and COVID-19 restricted follow-up surveys. A larger nonadopter sample would improve robustness. We note that the adoption literature has been dominated by economists; future research should integrate insights from agricultural engineering and hydrology to better capture technical and hydrological constraints.

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5 | Conclusion

The study examined the factors influencing MI adoption and adoption intensity in four major states in India using a Heckman two-stage regression. It identified information dissemination, land fragmentation, cultivation of high-value MI amenable crops, landholding size, irrigation sources, damage from animals and access to subsidies as important determinants of adoption. The results confirm that farmers find MI to be both agronomically and economically beneficial. Reliable information on the technology, subsidies and after-sale services empowers farmers to adopt, reinforcing the importance of informed decision-making. Adopters face significant damage from wild and stray animals impacting both irrigation technology and crops, underscoring the need for policies such as crop insurance, live wire fencing and rodent management extension. Improved institutional support, hassle-free subsidies and credit access could significantly boost adoption.

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Conflicts of Interest

The study was supported by Ministry of Agriculture and Farmers Welfare of the Government of India through the Centre for Management in Agriculture at the Indian Institute of Management, Ahmedabad. We declare that there were no competing financial or personal interests that influenced the study.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author(s) used *LanguageTool*, an AI grammar, style and spell checker tool to improve the readability and ensure coherence. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Description of continuous variables with referred work. **Table S2:** Description of categorical variables with referred work. **Table S3:** Heckman two-stage regression results for the adoption and intensity of micro-irrigation.

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